

Syntax Analyzer (Parsing)

Top-Down
Parsing

LL(1)
Grammar

① Recursive Descent
Parsing

② Predictive Parsing

LL(1)

{ Classroom : Slide ☒ }

Bottom - up

Parsing (aka Shift-reduced
Parsing)

① LR(0)

② LR(1) / CLR(1)

③ LALR(1)

L = Left to Right

R = Rightmost Derivation

(K) → look ahead

↳ number of input to be
parsed at once.

Bottom Up parsing

Shift reduce parsing

$S \rightarrow aTR_e$

$T \rightarrow Tbc \mid b$

$R \rightarrow d$

Input: a b b c d e

Step-1: Shift → a

a

Remaining input: b b c d e

Step-2:

shift $\rightarrow$ b
 $\begin{bmatrix} b \\ a \end{bmatrix}$
Remaining input: bcde

Step-3:

Reduce by $T \rightarrow b$
 $\begin{bmatrix} T \\ a \end{bmatrix}$
Remaining input: bcde

Step-4:

shift $\rightarrow$ b
 $\begin{bmatrix} b \\ T \\ a \end{bmatrix}$
Remaining input: cde

Step-5:

shift $\rightarrow$ c
 $\begin{bmatrix} c \\ b \\ T \\ a \end{bmatrix}$
Remaining input: de

Step-6:

Reduce by $T \rightarrow Tbc$
 $\begin{bmatrix} T \\ a \end{bmatrix}$
Remaining input: de

Step-9:

shift $\rightarrow$ e
 $\begin{bmatrix} R \\ T \\ a \end{bmatrix}$
Remaining input:

Step-7:

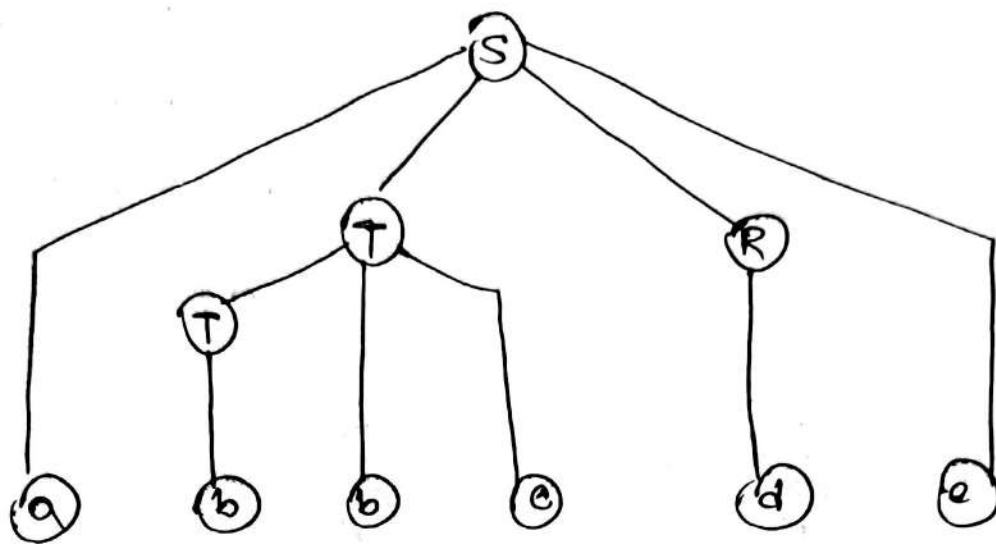
shift $\rightarrow$ d
 $\begin{bmatrix} d \\ T \\ a \end{bmatrix}$
Remaining input: e

Step-10:

Reduce by
 $\begin{bmatrix} S \end{bmatrix}$ $S \rightarrow aTRé$

Step-8:

Reduce by $R \rightarrow d$
 $\begin{bmatrix} R \\ T \\ a \end{bmatrix}$
Remaining input: e



2nd Rule

$E \rightarrow E + T$

$E \rightarrow T$

$T \rightarrow T * F$

$T \rightarrow F$

$F \rightarrow (E)$

$F \rightarrow id$

Operations

→ shift: push to stack

→ reduce: group as non-terminals

Input: id+id

Stack	Input	Action
\$	id+id \$	shift → id
\$id	+id \$	Reduce by $F \rightarrow id$
\$F	+id \$	Reduce by $T \rightarrow F$
\$T	+id \$	Reduce by $E \rightarrow T$
\$E	+id \$	shift → +
\$E+	id \$	shift → id

$\$E + id$

$\$E + F$

$\$E + T$

$\$E$

$\$$

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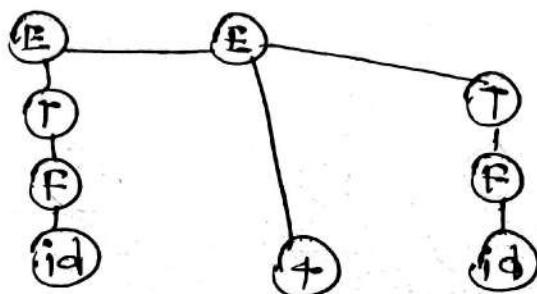
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Reduce $F \rightarrow id$

Reduce $T \rightarrow F$

Reduce by $E \rightarrow E + T$

Accept



$E \rightarrow 1E1$

$E \rightarrow 2E2$

$E \rightarrow 3$

Input : 12321

Step-1: Shift 1

$\begin{bmatrix} 1 \end{bmatrix}$ Remaining input : 2321

Step-2: shift 2

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ Remaining input : 321

Step-3: shift 3

$\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ Remaining input : 21

Step-4: Reduce by $E \rightarrow 3$

$\begin{bmatrix} E \\ 2 \\ 1 \end{bmatrix}$ Remaining input : 21

Step-5: Shift - 2

$\begin{bmatrix} 2 \\ E \\ 2 \\ 1 \end{bmatrix}$ Remaining input : 1

Step-6: Reduce by $E \rightarrow 2E2$

$\begin{bmatrix} E \\ E \\ 1 \end{bmatrix}$ Remaining input : 1

Step-7: Shift 1

$\begin{bmatrix} E \\ E \\ 1 \end{bmatrix}$ Remaining input : -

Step-8: Reduce by $E \rightarrow 1E1$

$\begin{bmatrix} E \end{bmatrix}$ Remaining input : -